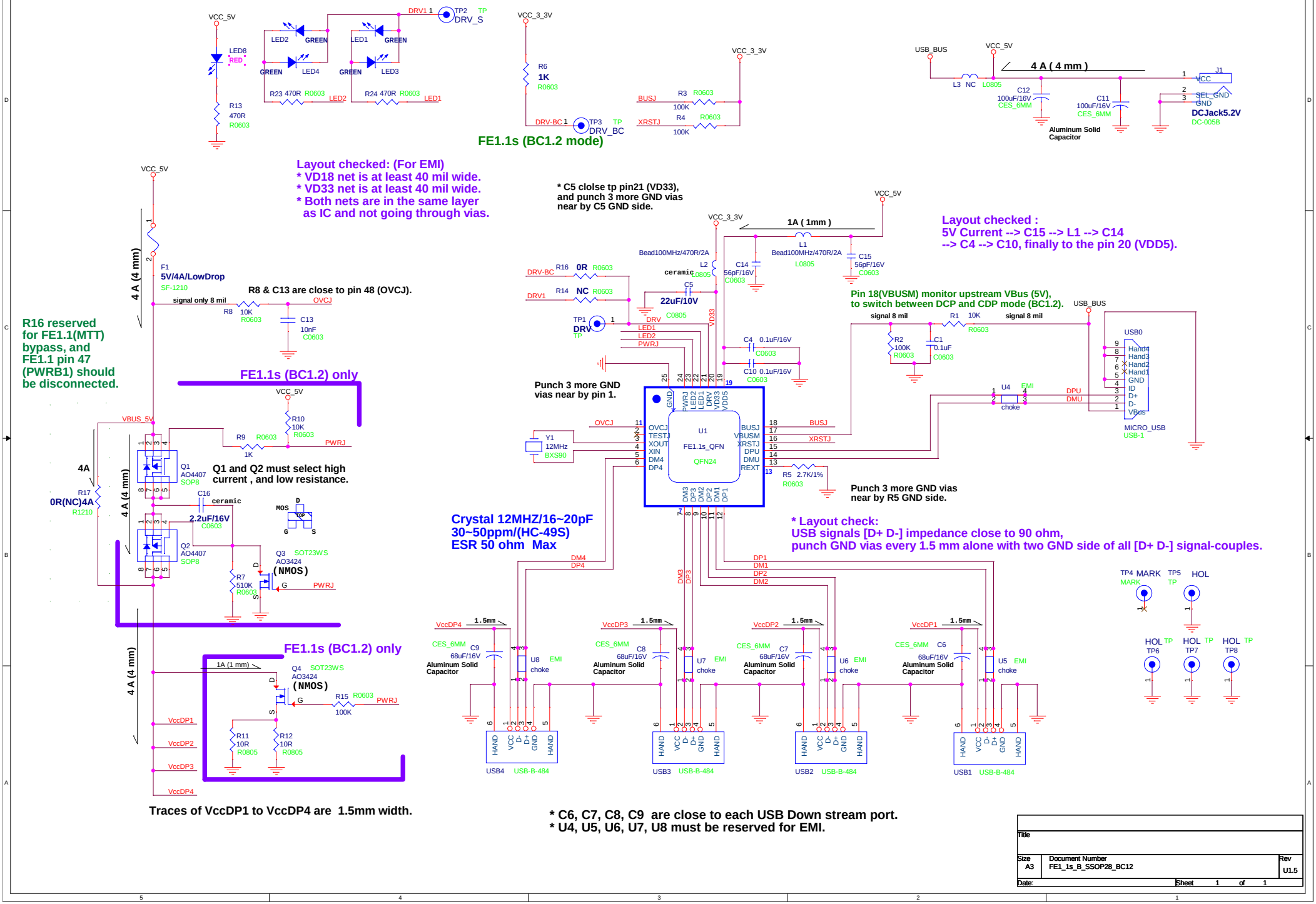
VCC_5V
LED8
RED
LED2
GREEN
LED1
GREEN
GREEN
LED4
GREEN
LED3
R23 470R R0603
LED2
R24 470R R0603
LED1
DRV1 1
TP2
TP
DRV_S
VCC_3_3V
R6
1K
R0603
DRV-BC 1
TP3
TP
DRV_BC
FE1.1s (BC1.2 mode)
VCC_3_3V
BUSJ
R3 R0603
100K
XRSTJ
R4 R0603
100K
USB_BUS
VCC_5V
4 A (4 mm)
L3 NC L0805
C12
100uF/16V
CES_6MM
C11
100uF/16V
CES_6MM
Aluminum Solid Capacitor
J1
VCC
SEL_GND
GND
DCJack5.2V
DC-005B
Layout checked: (For EMI)
* VD18 net is at least 40 mil wide.
* VD33 net is at least 40 mil wide.
* Both nets are in the same layer as IC and not going through vias.
* C5 clolse tp pin21 (VD33), and punch 3 more GND vias near by C5 GND side.
VCC_5V
4 A (4 mm)
F1
5V/4A/LowDrop
SF-1210
signal only 8 mil
R8 10K R0603
OVCJ
C13
10nF
C0603
R8 & C13 are close to pin 48 (OVCJ).
R16 reserved for FE1.1(MTT) bypass, and FE1.1 pin 47 (PWRB1) should be disconnected.
FE1.1s (BC1.2) only
VCC_5V
R10
10K
R0603
R9 R0603
1K
PWRJ
VBUS_5V
4A
R17
0R(NC)4A
R1210
4 A (4 mm)
Q1
AO4407
SOP8
Q1 and Q2 must select high current , and low resistance.
C16
ceramic
2.2uF/16V
C0603
Q2
AO4407
SOP8
MOS
top
D
G
S
Q3
SOT23WS
AO3424
(NMOS)
R7
510K
R0603
PWRJ
FE1.1s (BC1.2) only
4 A (4 mm)
1A (1 mm)
Q4
SOT23WS
AO3424
(NMOS)
R15 R0603
100K
PWRJ
VccDP1
VccDP2
VccDP3
VccDP4
R11
10R
R0805
R12
10R
R0805
Traces of VccDP1 to VccDP4 are 1.5mm width.
VCC_5V
VCC_3_3V
1A (1mm)
L1
Bead100MHz/470R/2A
L0805
C15
56pF/16V
C0603
C14
56pF/16V
C0603
Bead100MHz/470R/2A
L2
ceramic
L0805
C5
22uF/10V
C0805
DRV-BC
R16 0R R0603
DRV1
R14 NC R0603
TP1
DRV
TP
DRV
LED1
LED2
PWRJ
VD33
C4 0.1uF/16V
C0603
C10 0.1uF/16V
C0603
Punch 3 more GND vias near by pin 1.
Y1
12MHz
BXS90
U1
FE1.1s_QFN
QFN24
OVCJ
TESTJ
XOUT
XIN
DM4
DP4
BUSJ
VBUSM
XRSTJ
DPU
DMU
REXT
R5 2.7K/1%
R0603
Punch 3 more GND vias near by R5 GND side.
Layout checked :
5V Current --> C15 --> L1 --> C14
--> C4 --> C10, finally to the pin 20 (VDD5).
Pin 18(VBUSM) monitor upstream VBus (5V), to switch between DCP and CDP mode (BC1.2).
signal 8 mil
R1 10K
R0603
signal 8 mil
R2 100K R0603
C1 0.1uF C0603
USB_BUS
USB0
Hand4
Hand3
Hand2
Hand1
GND
ID
D+
D-
VBus
MICRO_USB
USB-1
U4
EMI
choke
DPU
DMU
Crystal 12MHZ/16~20pF
30~50ppm/(HC-49S)
ESR 50 ohm Max
* Layout check:
USB signals [D+ D-] impedance close to 90 ohm,
punch GND vias every 1.5 mm alone with two GND side of all [D+ D-] signal-couples.
DM4
DP4
DP1
DM1
DP2
DM2
DM3
DP3
VccDP4
1.5mm
C9
68uF/16V
CES_6MM
Aluminum Solid Capacitor
U8
EMI
choke
VccDP3
1.5mm
C8
68uF/16V
CES_6MM
Aluminum Solid Capacitor
U7
EMI
choke
VccDP2
1.5mm
C7
68uF/16V
CES_6MM
Aluminum Solid Capacitor
U6
EMI
choke
VccDP1
1.5mm
C6
68uF/16V
CES_6MM
Aluminum Solid Capacitor
U5
EMI
choke
HAND
VCC
D-
D+
GND
HAND
USB4 USB-B-484
USB3 USB-B-484
USB2 USB-B-484
USB1 USB-B-484
TP4 MARK
MARK
TP5 HOL
TP
HOL TP6
HOL TP7
HOL TP8
* C6, C7, C8, C9 are close to each USB Down stream port.
* U4, U5, U6, U7, U8 must be reserved for EMI.
Title
Size A3
Document Number FE1_1s_B_SSOP28_BC12
Rev U1.5
Date:
Sheet 1 of 1